



OPERATION Department
SIMHADRI

Note No :1



Ref:SMTTP/OPERATION/2023-24/APH Fire/450176

Date:

04/09/2023(00:58:27)

Subject:REVISION:5 OF LMI ON PREVENTION OF AIR PRE HEATERS FIRES

Ref:SMTTP/OPERATION/2023-24/APH Fire/450176

Date:04/09/2023

Subject:REVISION:5 OF LMI ON PREVENTION OF AIR PRE-HEATERS FIRES

Respected Sir,

This LMI on "prevention of air pre-heaters fire" is revised inline with Corporate Operation Directive "COS-ISO-00-OD/OPS/SYST/004 Rev. No:5 Date: August 2023"

Revision details:

- Point no. 9 "Degree of Superheat of S/B Steam" added in Protocol Annexure-IV.
- Annexure-VIII(a) added for Front and rear fired boilers. **(Not Applicable to Simhadri ST-I and ST-II)**
- Reference OIN-OPS-MECH-023 has been revised as OGN-OPS-MECH-055 in 4.5.2 f note.

The revised LMI and OD documents are attached.

Submitted for approval of competent authority.

Submitted by:

अघारा उदय कुमार सावजीभाई - उप प्रबंधक

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On:04/09/2023(00:58:27) **Ref:**SMTTPP/OPERATION/2023-24/APH Fire/450176

Note No:2

As discussed, for review please.

Submitted by:

मलय कांति जाना - अपर महाप्रबंधक

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On:04/09/2023(18:50:23) **Ref:**SMTTPP/OPERATION/2023-24/APH Fire/450176

Note No:3

Respected Sir,

This LMI on "Prevention of Air Pre-heaters fire" is revised inline with Corporate Operation Directive "COS-ISO-00-OD/OPS/SYST/004 Rev. No:5, Date: August 2023"

Following necessary Revision done in LMI:

- (a) Point no. 9 "Degree of Superheat of S/B Steam" added in Protocol Annexure-IV.
- (b) Annexure-VIII(a) added for Front and rear fired boilers. **(Not applicable for Simhadri ST-I and ST-II)**
- (c) Reference OIN-OPS-MECH-023 has been revised as OGN-OPS-MECH-055 in 4.5.1 f note.

Point (B) is not added in LMI as it is not applicable for Simhadri ST-I and ST-II.

The Revised LMI and OD documents are attached.

Submitted for approval of competent authority

Submitted by:

अघारा उदय कुमार सावजीभाई - उप प्रबंधक

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On:05/09/2023(13:23:29) **Ref:**SMTTPP/OPERATION/2023-24/APH Fire/450176

Note No:4

Proposed LMI may please be reviewed W.R.T. OGN.

Submitted by:

मलय कांति जाना - अपर महाप्रबंधक

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On:08/09/2023(18:30:06) **Ref:**SMTTP/OPERATION/2023-24/APH Fire/450176

Note No:5

Checked and found in order.

Submitted by:

सम्राट बनिक - प्रबंधक

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On:08/09/2023(18:53:36) **Ref:**SMTTP/OPERATION/2023-24/APH Fire/450176

Note No:6

LMI Custodian/initiator may be asked to prepare and present the Gist of this LMI in DPM for the benefit of the sectional heads and clearly spell out the responsibilities of all sections as per this LMI to the fora

The presentation may be completed at an earliest day and forward the LMI for further processing

Submitted by:

टी.आर. राजगोपालन् - महाप्रबंधक

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On:09/09/2023(09:46:07) **Ref:**SMTTP/OPERATION/2023-24/APH Fire/450176

Note No:7

Do needful as per noting.

Submitted by:

मलय कान्ति जाना - अपर महाप्रबंधक

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On:09/09/2023(18:55:59) **Ref:**SMTTP/OPERATION/2023-24/APH Fire/450176

Note No:8

Presentation Done. Kindly review.

Submitted by:

अघारा उदय कुमार सावजीभाई - उप प्रबंधक

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On:07/11/2023(16:29:08) **Ref:**SMTTP/OPERATION/2023-24/APH Fire/450176

Note No:9

Necessary approval may please be accorded.

Submitted by:

मलय कांति जाना - अपर महाप्रबंधक

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On:07/11/2023(17:05:18) **Ref:**SMTTP/OPERATION/2023-24/APH Fire/450176

Note No:10

Submitted by:

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On:10/11/2023(18:07:27) **Ref:**SMTTP/OPERATION/2023-24/APH Fire/450176

Note No:11

Reviewed.

Submitted by:

मकाला नागेश्वर राव - उप महाप्रबंधक

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On:11/11/2023(10:21:05) **Ref:**SMTTP/OPERATION/2023-24/APH Fire/450176

Note No:12

Submitted by:

धर्मेंद्र कुमार तोमर - अपर महाप्रबंधक

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On:11/11/2023(11:23:15) **Ref:**SMTTP/OPERATION/2023-24/APH Fire/450176

Note No:13

Maintence of APH Main Motor and Soot blower Motor will be carried out by EMD after taking PTW.

Submitted by:

कृष्णा बीसेट्टी - उप महाप्रबंधक

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On:13/11/2023(16:53:14) **Ref:**SMTTP/OPERATION/2023-24/APH Fire/450176

Note No:14

Reviewed. May please be approved.

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

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On:14/11/2023(10:18:08) **Ref:**SMTTP/OPERATION/2023-24/APH Fire/450176

Note No:15

Reviewed and forwarded for further approval.

Submitted by:

जी एच राव - अपर महाप्रबंधक

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On:14/11/2023(22:32:11) **Ref:**SMTTP/OPERATION/2023-24/APH Fire/450176

Note No:16

Submitted by:

ऊमा शंकर वर्मा - अपर महाप्रबंधक

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On:21/11/2023(17:49:59) **Ref:**SMTTP/OPERATION/2023-24/APH Fire/450176

Note No:17

Submitted by:

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On:23/11/2023(13:44:45) **Ref:**SMTTP/OPERATION/2023-24/APH Fire/450176

Note No:18

Forwarded for approval.

Submitted by:

राजेश चावला - अपर महाप्रबंधक

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On:23/11/2023(16:18:16) **Ref:**SMTTP/OPERATION/2023-24/APH Fire/450176

Note No:19

May please be accorded approval.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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On:27/11/2023(16:43:57) **Ref:**SMTTP/OPERATION/2023-24/APH Fire/450176

Note No:20

May please be approved.

Submitted by:

धुरजति प्रसाद पात्रा - महाप्रबंधक

Dhurjati Prasad Patra - 004573 (GENERAL MANAGER ,O & M)
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On:27/11/2023(20:01:37) **Ref:**SMTTP/OPERATION/2023-24/APH Fire/450176

Note No:21

Approved & Submitted by:

संजय कुमार सिन्हा - महाप्रबंधक

Sanjay Kumar Sinha - 005185 (GENERAL MANAGER ,HEAD OF PROJECT)
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On:28/11/2023(09:52:49) **Ref:**SMTTP/OPERATION/2023-24/APH Fire/450176



NTPC LIMITED

Simhadri Super Thermal Power Station

LOCATION MANAGEMENT INSTRUCTION

LMI/OD/OPS/SYST/004

Rev: 05 Date: AUGUST 2023

PREVENTION OF AIR PRE HEATERS FIRES

Approved for
Implementation by
HOP (Simhadri)

Date:

LMI	<i>PREVENTION OF AIR PRE-HEATERS FIRES</i>		
LMI No	LMI/OD/OPS/SYST/004		
Issue No	1	Revision: 5	Dt: 02.09.2023

	Name	Designation	Signature
Prepared by:	AGHARA UDAYKUMAR SAVJIBHAI	Dy. Manager (O)	
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Rechecked by:	T.R.RAJAGOPALAN	GM (O)	
Checked by	D.K. TOMAR	AGM(EMD)	
Checked by	OM PRAKASH	AGM (BMD)	
Checked by	G. HARI NARAYANA RAO	AGM(C&I)	
Checked by	UMA SHANKER VERMA	AGM(EEMG)	
Checked by	RAJESH CHAWLA	AGM(MTP)	
Rechecked by	MATHEW EIPE KOVOOR	GM (Maint)	
Reviewed by:	D.P.PATRA	GM (O&M)	
Approved by	S.K.SINHA	HOP (Simhadri)	

CONTENTS

Sl.No.	Description	Page No
1.0	INTRODUCTION	4
2.0	SUPERSEDED DOCUMENTS & REVISION DETAILS	4
3.0	SCOPE	4
4.0	GUIDANCE	4
4.1	MECHANISM OF AIR HEATER FIRES	4
4.2	CAUSES OF AIR HEATER FIRES	5
4.3	DETECTION OF AIR HEATER FIRES	5
4.4	DEALING WITH AIR HEATER FIRES	6
4.5	THE MEANS OF PREVENTING AIR HEATERS FIRES	6
5	IMPLEMENTATION & RECORDS	9
6	REFERENCE DOCUMENT	10
7	REVIEW	10
8	DISTRIBUTION LIST	10
Annexure	1 Protocol for Air Heater Cleanliness 2 Protocol for Air Heater Fire Water 3 Protocol for Air Heater Oil Carry Over Probes (OCP) 4 Protocol for Air Heater Soot blower 5 Protections and Interlocks Air Heater Motor Auto Start Checking 6 Air Heater Auxiliaries Availability 7 Protocol for Oil Trip Valve 8 Protocol for Oil Burner Valve 9 Protocol for Oil Burners 10 500 MW Boilers Cold Start-Up Check List(Post O/H) 11 Boiler Start Up Log	11-25

PREVENTION OF AIR HEATER FIRES

1.0	INTRODUCTION
	In coal fired power plants, air heaters perform a major role by improving the boiler efficiency, improving combustion and reducing pollution by lowering the exit gas temperature. During start-ups and shut-down and sometimes even in boxed up (boiler) conditions, if proper care is not taken, air heater fires can occur which in turn will cause severe damage to plant and equipment. The subsequent downtime for repair and loss of availability can be very high. This LMI draws the attention of Station O&M personnel to the danger of serious fires in air pre-heaters and identifies the possible means of preventing them or reducing the damage in the unfortunate event of an air heater fire
2.0	SUPERSEDED DOCUMENTS & REVISION NOTES
	Superseded: LMI/OD/OPS/SYST/004, REV No 4, Dated 26.07.2021: Prevention of Air Heater Fires Revision: a) Point no. 9 “Degree of Superheat of S/B Steam” added in Protocol Annexure-IV. b) Annexure-VIII(a) added for Front and rear fired boilers. (Not Applicable to Simhadri ST-I and ST-II) c) Reference OIN-OPS-MECH-023 has been revised as OGN-OPS-MECH-055 in 4.5.1 f note.
3.0	SCOPE:
	This LMI document is applicable for 4 X 500 MW Units of Simhadri STPS, Stage-I & II units where Primary and Secondary air preheating is done by Regenerative type of air-heaters.
4.0	GUIDANCE
4.1	MECHANISM OF AIR HEATER FIRES
	Air heater being the coolest part of the heat exchange system of a boiler, un burnt fuel tends to deposit on air heater surface. Regenerative Air heaters installed are particularly vulnerable to fires because of high surface to weight ratio, compared with other heat exchange surfaces in the boiler. During light-up or shut-down process involving oil firing, if oil deposition takes place on the air heater heating elements and the ignition temperature (250 to 350C) is reached, air heater fire starts, as adequate air is available for supporting combustion. This mainly happens during a cold start-up due to poor combustion of oil. Un burnt or partially burnt oil will be carried over from the furnace and it gets deposited on the heating elements surface. Oil carryover can also take place due to passing of oil valves or leaking oil hoses. There had also been cases of fires in air heaters after shut-down after a prolonged period of oil firing. Initial ignition starts in a small area of the deposit. Most of the heat generated is absorbed by the metal of heat transfer element nearby. The temperature build up is slow if, the fire is detected at this time, it can be controlled easily. If this ignited deposit remains undetected, it will continue to generate heat and gradually the air heater heating elements start igniting, as the metal ignition temperature (about 1200 C) is reached. The metal fire is self-sustaining and it spreads very fast, burning the adjacent sheets. The effort required for extinguishing this fire will be very high. If undetected for about 2 hours it can damage the elements to the extent of about 50 %.

4.2	CAUSES OF AIR HEATER FIRES:
4.2.1	Air heater fires are mainly caused by the deposition of un burnt fuel on the air heater heat exchange surfaces, where it can be subject to significant temperatures and be supplied with adequate air to support combustion
4.2.2	The fires that have occurred on air heaters have been associated with prolonged oil firing of the boiler, added with poor combustion and delayed / ineffective soot blowing.
4.2.3	Most of the air heater fires have occurred in air heaters during cold start up after replacement of heating elements. This stresses the need to properly preserve and store the baskets after receipt at site, such that the baskets remain clean.
4.2.4	The major reasons leading to air heater fires can thus be summarized as follows: a) Carryover of un-burnt oil through the boiler to the air heater due to delayed ignition. b) Carryover of un-burnt oil due to passing of burner valves or leakage in the oil hose/tip. c) Carryover of un-burnt oil or soot due to improper combustion arising from: • Inadequate oil pre-heating (in case of HFO). • Incorrect burner fuel oil pressures. • Incorrect atomizing medium pressure. • Defective burners. • Inadequate combustion air supply/ temperature • Inadequate ignition energy (mainly for HFO burners) • Defective scanners d) Non removal of combustible material before attaining the ignition temperature by means of on line soot blowing.
4.3	DETECTION OF AIR HEATER FIRES
	Early detection of air heater fire will help in reducing the damage and down time. An air heater fire is indicated by one or more of the following symptoms. a) Abnormal or sudden rise in temperature, mainly, gas / air outlet temperature. Once the fire is established, all the terminal temperatures may go out of scale. b) An alarm indicating 'Air heater gas outlet / air outlet temp high c) Stoppage or jamming of the air heater (air heater rotor stoppage alarm) or hunting in the current d) Visual indications at local, mainly through inspection port (other than smoke outside as it will be sucked by the ID fan) e) Sudden jamming of the dampers below.

4.4	DEALING WITH AIR HEATER FIRES
	Any sign of an air heater fire should be dealt with very seriously. On receiving a suspicion of air heater fire, start air heater soot blowers immediately (if not in progress already) and inspect the air heater locally for any abnormality. If there is no sign of improvement and fire is confirmed, then proceed as follows:- a) Trip the boiler and drought plant and close all the associated dampers, leaving, the air heater running b) Open fire water and water washing valves for extinguishing the fire c) Intimate fire tender. d) Admit fire water in the air side also, through the header, if provided; or with the help of fire tender through the manhole. e) Continue to spray water until the rotor is completely cooled. Ensure that water is free to escape through drains/ hoppers f) Keep the air heater running and open the inspection doors when safe to examine the air heater g) Inspect the other air heaters also.
	Responsibility:UCE
<u>Note:</u>	- While admitting fire water, bottom duct/ hopper drains / manholes should be opened. Dampers also can be slightly opened for the escape of water and steam. - The same procedure mentioned above is to be adopted, if the air heater has become standstill also (jammed). - Soot blowing operation is to be suspended during fire fighting. - Copious amount of water only and not other extinguishers like CO₂, Halon etc. are ineffective during air heater fire.
4.5	THE MEANS OF PREVENTING AIR HEATERS FIRES
	Air heater fires can be avoided if enough care is taken during overhaul, re-commissioning and operation of the plant. The precautions to be taken at various O&M stages are given below:
4.5.1	DURING OVERHAUL:
	a) Old Heating elements are to be cleaned thoroughly by HP water jet during overhaul Resp: MM-RM b) Healthiness of oil firing system (oil guns, ignitors, scanners etc.) shall be ensured Resp: Opn, C&I, PP c) Availability of Fire Detection System shall be ensured Resp: Opn, C&I d) Availability of Soot Blowing System shall be ensured Resp: Opn, RM, EM, C&I e) Healthiness of firefighting/ water washing headers and nozzles shall be ensured during overhaul. Resp: Opn, RM f) The scope of work for air heater shall cover air heater auxiliaries also, which are meant for monitoring/ prevention of fire hazards (soot blower, fire detection system, inspection port, oil carry over probe, fire-fighting system etc.) Note: For more information on air heater maintenance, refer OGN-OPS-MECH-055; Operation and Maintenance Practices of Air Pre-heaters. Resp: Opn, RM, EM, C&I g) Availability of fire hydrants near air heater must be ensured Resp: MM-OS

4.5.2	DURING RECOMMISSIONING (PRE-LIGHT UP)
	a) All men & material should be removed before air pre heater is boxed up. The necessary protocols for the same have to be filled up before lighting up the boiler. This ensures proper cleanliness of the APH before light up.
	Resp: MM-RM
	b) Air heater dampers shall be commissioned and made available for operation.
	Resp: Opn, RM, EM, C&I
	c) Air heater oil carry over probes and inspection ports with lights shall be made available for operation. Oil carry over probe to be made free to operate (insert and removal) and cooling water line to be charged.
	Resp: RM, EM
	d) Air heater fire water shall be charged up to the individual valve. During commissioning fire water shall be charged and spray checking to be done, any nozzle chocking also to be cleared.
	Resp: Opn, RM
	e) Air heater soot blowers shall be commissioned and kept ready for operation. During commissioning, lance travel, coverage, steam pressure, valve operation, electrical operation, healthiness of pressure gauge etc are to be ensured
	Resp: Opn, RM, EM, C&I
	f) All oil burner valves shall be checked for passing and any defective one should be rectified
	Resp: Opn, PP
	g) Particular attention shall be given to the cleanliness and maintenance of oil burners. Blockage, wear or maladjustment of these precision installations will all play a part in reducing combustion performance. Guns shall be flushed thoroughly and any leakage from the sprayer or flexible hose should be attended
	Resp: PP
	h) All instruments related to combustion and air heaters must be ensured to be in perfect working condition by proper calibration/ validation (indicators, recorders etc in control room and local gauges).
	Resp: C&I
	i) Fire detection system shall be available when the air heater is in service, particularly during start - up activities
	Resp: C&I
	j) Air heater interlocks /protections and the functioning of alarms shall be checked before taking the air heater into service
	Resp: Opn, C&I
4.5.3	DURING START UP
	During start up as a general rule OEM recommendations / integrated start up procedures are to be followed. The following points are stressed in addition
	a) Necessary protocols, covering the aspects mentioned in 4.5.2 shall be signed with the concerned maintenance departments, before boiler light up (Annexure I to IX).
	Resp: Opn, MM, EM, C&I
	b) Boiler cold start up check list should be signed and the deviations shall be approved by the competent authority (Annexure X).
	c) During start up, keep all the air heaters in service, irrespective of the number of fans

	being started
	Resp: OPN
	d) Keep the air and flue gas dampers of the idle air heaters closed (Where dampers are not available, keep both the ID, FD fans in service).
	Resp: OPN
	e) Charge SCAPHs as soon as the fans are started and maintain the recommended air temp (about 70°C)
	f) For cold start up after heating elements replacement, after attaining the SCAPH outlet temp, the elements shall be dried for about two hour before start of oil firing.
	Resp: OPN
	g) Ensure correct air flow at the time of purging and light up
	Resp: OPN
	h) Maintain proper atomizing medium pressure and oil pressure at the time of light up.
	Resp: OPN
	i) Maintain the recommended minimum oil temperature (about 100°C for HFO).
	Resp: OPN
	j) During light up, check for proper combustion by observing the flame at local, exit of the furnace section and the stack. Take corrective action in case of unstable flame or black or white smoke leaving the chimney
	Resp: OPN
	k) Ensure that individual flame scanners are sensing properly (confirmed by scanner calibration before light up) and flame protection is in service
	Resp: OPN
	l) Secondary air dampers of all oil guns in service shall be ensured open.
	Resp: OPN
	m) If the flame of any oil burner is not proper, the burner should be withdrawn immediately and should not be taken into service till the defects are rectified. Burners, which are not proving shall not be tried repeatedly.
	Resp: OPN
	n) Air heater soot blowing shall be started immediately after boiler light up. Blowing should be done with steam at proper pressure and temperature continuously throughout the period when the oil guns are in service, (including the idle air heater). During light up, if availability of the soot blower is lost, the boiler should be boxed up immediately
	Resp: OPN
	o) Oil carry over probes shall be inspected regularly during oil firing. If oil carry over is observed then oil flame should be checked and oil guns having improper flame should be withdrawn immediately.
	Resp: OPN
	p) Air heaters shall be inspected locally through the inspection port regularly
	Resp: OPN
	q) Increase of firing rate for raising steam parameters shall be as per the rates prescribed under start up procedures as supplied by the OEM / OES.
	Resp: OPN
	r) Heavy oil firing shall be started gradually only after reaching a furnace temperature of

	300°C
	Resp:OPN
	s) While the boiler and air heaters are in service, all the instruments related to combustion and air preheaters should be observed carefully and any sign of error in the instrument should be attended immediately.
	Resp:OPN
	t) A close watch should be kept on the terminal temperatures of air heaters. Any unusual increase in one or more of these should be investigated immediately.
	Resp:OPN
	u) Boiler start up parameters are to be recorded regularly (refer boiler start up log Annexure XI).
	Resp:OPN
	v)Monitoring as above shall be done regularly till oil support is completely removed (and temperatures have stabilized) but more vigorously in the flue gas inlet temp range of 200-250°C
	Resp:OPN
	w)Air heater soot blowing shall also be carried out while shutting down the boiler after oil firing
	Resp:OPN
4.5.4	DURING SHUT DOWN:
	There is some amount of risk of air heater fire after shutting down the boiler after a prolonged period of oil firing. So monitoring of the air heater parameters shall be continued in a shut down unit also.
	a)After the boiler is tripped and combustion chamber has been purged, air heaters shall be kept in service till completely cooled
	Resp:OPN
	b) The air heater terminal temperatures shall be monitored regularly when the boiler is boxed up till the air heater cools down. Any increase in any of the temperatures shall be investigated immediately
	Resp:OPN
5	IMPLEMENTATION & RECORDS
	The following records shall be maintained in connection with this LMI :
	a)Signed Protocols (air heater cleanliness, cleaning of heating elements, air heater firewater, air heater oil carry over probes, air heater soot blower, air heater Protections & Interlocks, air heater aux. availability, oil burners, burner valves, Oil trip valve etc)
	b)Boiler cold start up checklist (signed)
	c)Boiler start up parameters
	Resp:OPN
Note:	Sample formats for protocols and checklists are attached with this LMI as annexure.

6.0	REFERENCE DOCUMENT:
	This LMI is based on Operation Directive "COS-ISO-00-OD/OPS/SYST/004 Rev. No.: 4 Date: June 2021"
7.0	REVIEW
	The HOP will be responsible for reviewing the LMI on a 2- yearly basis or as required
8.0	DISTRIBUTION LIST:
	1. HOD(OPN) 2. SHIFT CHARGE ENGINEER 3. ALL DEPTT OF O&M THROUGH LAN

ANNEXURE-I

PROTOCOL FOR AIR HEATER CLEANLINESS

UNIT:

DATE:

AIR HEATER:

ACTIVITY: APTS INTERNAL INSPECTION

SCOPE: ALL AIR PREHEATERS

SL.NO	DESCRIPTION	STATUS	REMARKS
1	ALL FOREIGN MATERIAL MATERIALS ARE REMOVED FROM THE AIR HEATER AND THE BASKETS ARE CLEAN		
2	AIRHEATER BASKETS ARE FREE FROM ASH PARTICLES		
3	AIRHEATER BASKETS ARE FREE FROM OIL TRACES AND OTHER COMBUSTIBLE MATERIAL.		

CHECKED & CERTIFIED BY	NAME	DESIGNATION	SIGNATURE
BMD			
RECOMMISSIONING			

ANNEXURE-II

PROTOCOL FOR APH FIRE WATER

UNIT:

DATE:

AIR HEATER:

ACTIVITY: CHARGING OF FIRE WATER TO APHS

SCOPE: APH FIRE WATER

SI No	DESCRIPTION	STATUS	REMARKS
1	FIRE WATER MAIN VALVE OPENED AND CHARGED UPTO INDIVIDUAL VALVES		
2	DRAIN AFTER INDIVIDUAL VALVES KEPT OPEN		
3	PASSING OF FIREWATER VALVES (INDIVIDUAL) CHECKED AND CLEARED		
4	NOZZLE CLEANLINESS ENSURED (BY PHYSICAL INSPECTION OR DURING WATER WASHING)		

CHECKED & CERTIFIED BY	NAME	DESIGNATION	SIGNATURE
BMD			
RECOMMISSIONING			

ANNEXURE-III

PROTOCOL FOR AIR PREHEATER OIL CARRY OVER PROBES (OCP)

UNIT:

DATE:

AIR HEATER:

ACTIVITY: RECOMMISSIONING OF OCP

SCOPE: ALL PRIMARY AND SECONDARY APH

SI No	DESCRIPTION	STATUS	REMARKS
1	LANCE HEALTHINESS		
2	FREENESS CHECKED AND PROBE INSERTED		
3	COOLING WATER CONNECTIONS		
4	WATER LEAKAGE		

NOTE: OCP is to be kept outside after completion of light up activities to prevent over heating/water leakages.

CHECKED & CERTIFIED BY	NAME	DESIGNATION	SIGNATURE
BMD			
RECOMMISSIONING			

ANNEXURE-IV

PROTOCOL FOR AIR PREHEATER SOOTBLOWER

UNIT:

DATE:

AIR HEATER:

ACTIVITY: RECOMMISSIONING OF APH SOOT BLOWERS

SCOPE: ALL APH SOOT BLOWERS (PRIMARY AND SECONDARY)

SL NO	DESCRIPTION	STATUS	REMARKS
1	LANCE MOVEMENT		
2	FREENESS		
3	COVERAGE (WITH STEAM)		
4	NOZZLE CONDITION		
5	DISTANCE FROM CASING		
6	STEAM LEAKAGE		
7	PUPPET VALVE OPERATION		
8	ELECTRICAL OPERATION		
9	DEGREE OF SUPERHEAT OF S/B STEAM		

CHECKED & CERTIFIED BY	NAME	DESIGNATION	SIGNATURE
BMD			
RECOMMISSIONING			

ANNEXURE-V

PROTECTIONS AND INTERLOCKS

AIR PREHEATER AIR MOTOR AUTO START CHECKING

UNIT:

DATE:

AIR HEATER:

SL.NO	DESCRIPTION	STATUS	REMARK
1	MAIN MOTOR TRIP TO AIR MOTOR AUTO START		
2	APH ELECTRICAL MOTOR TRIP ANNOUNCEMENT IN UCB		
3	AIR MOTOR RUNNING INDICATION IN UCB		
4	AIR MOTOR RUNNING ANNOUNCEMENT IN UCB		
5	AIR PRE HEATER STANDSTILL ANNOUNCEMENT		

CHECKED & CERTIFIED BY	NAME	DESIGNATION	SIGNATURE
BMD			
RECOMMISSIONING			

ANNEXURE-VI

AIR PREHEATER AUXILIARIES AVAILABILITY

UNIT:

DATE:

AIR HEATER:

SL.NO	DESCRIPTION	STATUS	REMARKS
1	APH SOOT BLOWER		
2	OIL CARRY OVER PROBE		
3	IRDS/ ALTERNATE FIRE SENSING MECHANISM		
4	FIRE WATER		
5	SUPPORT BEARING LOPS		
6	GUIDE BEARING LOPS		
7	INSPECTION DOOR & LIGHT		
8	GAS/AIR DAMPERS WITH REMOTE OPERATION		

CHECKED & CERTIFIED BY	NAME	DESIGNATION	SIGNATURE
BMD			
RECOMMISSIONING			
EMD			
C&I			

ANNEXURE-VII

PROTOCOL FOR OIL TRIP VALVE

UNIT:

DATE:

ACTIVITY: CHECKING OF THE TRIP VALVE FOR ANY PASSING KEEPING THE BURNER OIL VALVE AND MANUAL VALVE OPEN

SCOPE: HFO TRIP VALVE & LDO TRIP VALVE

SL.NO	DESCRIPTION	STATUS	REMARKS
1	HFO TRIP VALVE PASSING CHECKING		
2	OIL PR BEFORE HFO TRIP VALVE		
3	LDO TRIP VALVE PASSING CHECKING		
4	OIL PR BEFORE LDO TRIP VALVE		

CHECKED & CERTIFIED BY	NAME	DESIGNATION	SIGNATURE
BMD			
RECOMMISSIONING			

ANNEXURE-VIII

PROTOCOL FOR OIL BURNER VALVE

UNIT:

DATE:

ACTIVITY: CHECKING OF BURNER VALVE PASSING BY KEEPING THE
MANUAL VALVE AND MAIN TRIP VALVE OPEN

SCOPE: BURNER VALVE OF IN DIFFERENT ELEVATION ALL CORNERS

1. OIL PRESSURE:

2. OIL TEMPERATURE:

FOR STAGE-I				
	CORNER-1	CORNER-2	CORNER-3	CORNER-4
AB OIL VALVE				
CD OIL VALVE				
EF OIL VALVE				
FG OIL VALVE				
HK OIL VALVE				
(FOR STAGE-II)				
AB OIL VALVE				
CD OIL VALVE				
EF OIL VALVE				
GH OIL VALVE				
JK OIL VALVE				

CHECKED & CERTIFIED BY	NAME	DESIGNATION	SIGNATURE
BMD			
RECOMMISSIONING			

ANNEXURE-IX

PROTOCOL FOR OIL BURNERS

UNIT:

DATE:

ACTIVITY: RECOMMISSIONING CHECKS OF OIL BURNERS

**SCOPE: ALL OIL BURNERS (AB, CD, EF, FG, HK-IN STAGE-I), (AB, CD, EF, GH, J K ---IN
STAGE-II)**

SL.NO	DESCRIPTION	STATUS	REMARKS
1	COMPLETION OF MECHANICAL WORKS ON BURNER FRONT		
2	C&I CLEARANCE		
3	GUNS AND IGNITORS ADVANCE AND RETRACT CHECKING		
4	FLUSHING OF OIL GUNS FOR CLEANING		
5	ANY HOSE LEAKAGES		
6	IGNITERS SPARK HEALTHINESS		
7	BURNER VALVES PASSING CHECKING		

CHECKED & CERTIFIED BY	NAME	DESIGNATION	SIGNATURE
BMD			
RECOMMISSIONING			
C&I			

ANNEXURE-X

COLD START-UP CHECK LIST (POST OVERHAUL)

UNIT:

DATE:

SL.NO	DESCRIPTION	RESP	STATUS	REMARKS
1	Men & material removed from furnace & second pass & manholes /peepholes closed.	PP		
2	All ducts cleared and manholes closed.	PP		
3	All works on pressure parts completed and boxed up	PP		
4	All safety valve gags and hydro-plugs removed	PP		
5	All available oil guns are flushed checked for any leakage, cleaned and clamped.	PP		
6	All wall blowers and LRSBS are retracted fully	PP		
7	All air pre-heater soot blowers checked & cleared for operation	RM		
8	Men and material cleared from ducts, dampers, fans and air preheaters.	RM		
9	All air pre-heaters fire water is charged upto individual valve	RM		
10	All air pre-heater oil carry over probes are fixed in position and cleared for operation.	RM		
11	Clearance given from mills, feeders & fuel	RM		
12	Men and material removed from bottom ash hopper, economizer hopper, air pre-heater hopper and boxed up.	AHM/PP		
13	Minimum two passes in ESP boxed up and cleared for operation.	AHM		
14	Clearance given from critical piping HP / LP bypass & main turbine side	TMD		
15	All instruments related to APHS and boiler combustion are working	C&I		
16	For all APHS, fire detection system is kept in service.	C&I		
17	All protections & interlocks of boiler (FSSS) are checked & kept in service	C&I		

18	All alarms set as per the design, tested and kept in service	C&I		
19	APH soot blowers are cleared for operation	C&I		
20	Bottom ash hopper and seal trough. Filled.	Opn		
21	Boiler is filled up to the normal level	Opn		
22	Burner front & air pre-heaters Fire water ensured.	Opn		
23	Burner front, air pre-heaters and Other boiler areas are accessible	Opn		
24	All boiler protections are checked and kept in service	Opn		
25	Availability of APH fire detection System (or an alternative arrangement), oil carry over probes, air Pre-heater dampers & air pre-heater Soot blowers ensured	Opn.		
26	Service air availability ensured for Air pre-heater and igniters	Opn.		
27	Main turbine is on barring gear	Opn		
28	Clearance obtained from Pressure. Parts Rotary Machines, Ash handling Maint. , Turbine Maint, Elect & C&I Maint. for light-up and all major PTWs are cancelled.	Opn		
29	Major deviations from standard. Check list noted and submitted for Approval by competent authority	Opn		
30	Ash evacuation system readiness	Oprn/ash handling		

CLEARANCES

SI No.	DEPARTMENT	SIGNATURE	REMARKS
1	Pressure Parts (PP)		
2	Rotating Machines (RM)		
3	Turbine Maintenance (TMD)		
4	C&I Maintenance (C&I)		
5	Ash Handling Maint. (AHM)		
6	Operation / Recommissioning		

DEVIATIONS:

Shift Charge Engineer

AGM (Opn.)

GM (Opn)

Head (O&M) :

ANNEXURE-XI

BOILER START UP LOG

UNIT:

DATE:

S.N.	ACTIVITY DESCRIPTION	PARAMETER		REMARKS
		Recommended*	Actual	
01	Initial Unit condition: Cold / Hot			
02	Drum Metal temp °C-L / R:			
03	Boiler Light up time, hrs:			
04	Number of guns proved (within 10 min): Type of oil (HSD/ HFO):			
05	APH soot blowing started at, hrs-			
06	Second elevation guns taken in service at, hrs:			
07**	Boiler drum / SH vents closed at, hrs: Drum pr. ksc:			
08**	SH / RH header drains closed at, hrs: Drum pr. ksc:			
09**	MS line charged at, hrs: Drum pr. ksc:			
10	HP bypass charged at, hrs: MS pr, ksc:			
11	LP bypass charged at, hrs: HRH pr, ksc:			
12	Coal firing started at, hrs: Guns in service at that time:			
13	Turbine rolling started at:			
14	Sequence / time of starting subsequent mills:			
15	Unit synchronising time:			
16	Oil support removed at, hrs: Load, MW: (Total oil cons.):			
17	Full load reached at, hrs:			
18	Total deviation in start up time, hrs			
19	Constraints faced, if any during light up and loading:			
20	Action plan to address the constraints:			

* Recommended as per integrated start up procedures.

** Applicable during cold start ups

BOILER START UP LOG: MAJOR

PARAMETERS UNIT:

DATE:

S.N.	Parameter	Unit	1	2	3	4	Full Load
1.	Time	Hr:mn					
2.	LDO Pr.	Ksc					
3.	Atm. Air Pr	Ksc					
4.	HFO Pr.	Ksc					
5.	HFO temp	°C					
6.	Atm. Stm. Pressure	Ksc					
7.	Scanner air Pr	mmwc					
8.	Total Air Flow	TPH					
9.	Furnace Draft	mmwc					
10.	Flue Gas O ₂	%					
11.	Drum Pr.	KSC					
12.	Drum level	MM					
13.	Drum metal temp (maxm)	°C					
14.	ID fan-A Vane position	%					
15.	ID fan-B Vane position	%					
16.	ID fan -A Speed	rpm					
17.	ID fan-B Speed	rpm					
18.	FD Fan –A Blade pitch position	%					
19.	FD Fan –B Blade pitch position	%					
20.	SAPH-A air outlet temp	°C					
21.	SAPH-A gas inlet temp	°C					
22.	SAPH-A gas outlet temp	°C					
23.	SAPH-B air outlet temp	°C					
24.	SAPH-B gas inlet temp	°C					
25.	SAPH-B gas outlet temp	°C					
26.	PAPH-A air outlet temp	°C					
27.	PAPH-A gas inlet temp	°C					
28.	PAPH-A gas outlet temp	°C					
29.	PAPH-B air outlet temp	°C					
30.	PAPH-B gas inlet temp	°C					
31.	PAPH-B gas outlet temp	°C					
32.	SH temp before attemp..LHS	°C					
33.	SH temp after attemp..LHS	°C					
34.	SH temp before attemp.RHS	°C					
35.	SH temp after attemp.RHS	°C					
36.	MS temp. at Blr O/L LHS	°C					
37.	MS temp. at Blr O/L RHS	°C					
38.	SH attemp. Flow LHS	TPH					

S.N.	Parameter	Unit	1	2	3	4	Full Load
39.	SH attemp. Flow RHS	TPH					
40.	RH temp before attemp..LHS	⁰ C					
41.	RH temp after attemp..LHS	⁰ C					
42.	RH temp before attemp..RHS	⁰ C					
43.	RH temp after attemp..RHS	⁰ C					
44.	HRH Temp. LHS (blr side)	⁰ C					
45.	HRH Temp. RHS (blr side)	⁰ C					
46.	RH attemp. Flow LHS	TPH					
47.	RH attemp. Flow RHS	TPH					
48.	CBD opening	%					
49.	Silica (drum)	ppm					
50.	FW temp. at economizer inlet	⁰ C					
51.	Feed water flow/MS flow	TPH					
52.	MS flow	TPH					
53.	PA Fan –A Blade pitch position	%					
54.	PA Fan –B Blade pitch position	%					
55.	PA header pressure	mmwc					
56.	Seal air header pressure	mmwc					
57.	Total coal flow	TPH					
58.	Max. SH metal temp.	⁰ C					
59.	Max. RH metal temp.	⁰ C					